

SEC P vs NP log 2026-04-25

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-25 00:39 UTC

SEC P vs NP — daily log 2026-04-25

2 cycles recorded

Finite Geometry Line Count Bounds EF Proof Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite geometry (projective planes over $GF(2)$) $\times$ Extended Frege proof size
- **Recorded:** 2026-04-25 00:09 UTC
- **Entry ID:** 10350f5e2ac2

Statement

For any tautology φ in 3-CNF, the number of lines in the projective plane $PG(2, 2^k)$ constructed from φ 's clauses is $\Theta(2^{\{n/2\}})$ if and only if the EF proof size of φ is $\Omega(n^2)$.

Rationale

Finite geometry provides a combinatorial framework to encode clause interactions, while EF proof size measures logical complexity. The projective plane's line count captures structural constraints that may correlate with proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 14, 'conjecture_holds': True}
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_holds': True}
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 5, 'conjecture_holds': True}
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 8, 'conjecture_holds': True}
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_holds': True}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
```

~^~^~^~^~^~^

KeyError: 'seed'

Judge reasoning

The test failed to find a counterexample (empty 'counterexample' field) and the small instance sizes ($n \leq 14$) cannot validate asymptotic claims about $\Omega(n^2)$ vs $\Theta(2^{\{n/2\}})$ growth rates. | next: Test with larger n values (e.g., $n \geq 20$) and verify asymptotic behavior using rigorous complexity analysis

Dimension of Variety from Tautology Ideal Bounds EF Proof Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic geometry (Krull dimension of polynomial ideals) $\times$ Extended Frege proof size
- **Recorded:** 2026-04-25 00:39 UTC
- **Entry ID:** aba72d8e95aa

Statement

The Krull dimension of the variety defined by the ideal generated by the clauses of a tautology is $\Theta(\log \text{ size of Extended Frege proof for that tautology})$.

Rationale

The Krull dimension reflects the intrinsic complexity of the system's solution space, which may impose structural constraints on the proof complexity in Extended Frege.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bb1d86ad.py", line 91, in <module>
    seeds = [int(arg) for arg in sys.argv[1:]] or [11, 23, 37, 53, 71]
               ^^^
```

NameError: name 'sys' is not defined. Did you forget to import 'sys'?

Judge reasoning

The test code failed to execute due to a missing 'sys' import, yielding no empirical results. Without execution, we cannot assess support or falsification. | next: Fix the test code by importing 'sys' and re-run the experiment with proper setup.